Segway

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1 Boundry Condition

$$\mathrm{E_K^{body}} = \frac{1}{2}\,\mathrm{m_b}\,(\dot{\mathrm{x}}_\mathrm{b}^2 + \dot{\mathrm{y}}_\mathrm{b}^2 + \dot{\mathrm{z}}_\mathrm{b}^2) + \frac{1}{2}\,\mathrm{I_b}\,\dot{\theta}_\mathrm{b}^2$$

$$\mathrm{E_K^{wheels}} = \frac{1}{2}\,\mathrm{m_w}\,(\dot{\mathrm{x}}_{\mathrm{w_{left}}}^2 + \dot{\mathrm{x}}_{\mathrm{w_{right}}}^2 + \dot{\mathrm{y}}_{\mathrm{w_{left}}}^2 + \dot{\mathrm{y}}_{\mathrm{w_{right}}}^2 + \dot{\mathrm{z}}_\mathrm{w}^2) + \frac{1}{2}\,\mathrm{I_w}\,(\dot{\theta}_{\mathrm{w_{left}}}^2 + \dot{\theta}_{\mathrm{w_{right}}}^2)$$

$$\mathrm{E_P^{body}} = \mathrm{m_b}\,\mathrm{g}\,(\mathrm{l_b}\cos(\theta_\mathrm{b}) + \mathrm{z_w})$$

$$\mathrm{E_P^{wheels}} = \mathrm{m_w}\,\mathrm{g}\,\mathrm{z_w}$$

$$\mathrm{x_b} = \mathrm{x_w} + \mathrm{x_0} + \mathrm{l_b}\sin(\theta_\mathrm{b})\cos(\varphi_\mathrm{b})$$

$$\mathrm{y_b} = \mathrm{y_w} + \mathrm{y_0} + \mathrm{l_b}\sin(\theta_\mathrm{b})\sin(\varphi_\mathrm{b})$$

$$\mathrm{z_b} = \mathrm{z_w} + \mathrm{z_0} + \mathrm{l_b}\cos(\theta_\mathrm{b})$$

$$\mathrm{x_{w_{left}}} = \mathrm{x_w} - \frac{\mathrm{L}}{2}\sin(\varphi_\mathrm{w})$$

$$\mathrm{y_{w_{left}}} = \mathrm{y_w} + \frac{\mathrm{L}}{2}\cos(\varphi_\mathrm{w})$$

$$\mathrm{x_{w_{right}}} = \mathrm{x_w} + \frac{\mathrm{L}}{2}\sin(\varphi_\mathrm{w})$$

$$\mathrm{y_{w_{right}}} = \mathrm{y_w} - \frac{\mathrm{L}}{2}\cos(\varphi_\mathrm{w})$$

$$\dot{\mathrm{x}}_{\mathrm{w}} = \frac{\mathrm{z}_{\mathrm{w}}}{2} \left(\dot{\theta}_{\mathrm{w}_{\mathrm{left}}} + \dot{\theta}_{\mathrm{w}_{\mathrm{right}}}\right) \cos(\varphi_{\mathrm{w}})$$

$$\dot{\mathrm{y}}_{\mathrm{w}} = \frac{\mathrm{z}_{\mathrm{w}}}{2} \left(\dot{\theta}_{\mathrm{w}_{\mathrm{left}}} + \dot{\theta}_{\mathrm{w}_{\mathrm{right}}}\right) \sin(\varphi_{\mathrm{w}})$$

$$\dot{\varphi}_{\mathrm{w}} = \frac{\mathrm{z}_{\mathrm{w}}}{\mathrm{L}} \left(\dot{\theta}_{\mathrm{w}_{\mathrm{right}}} - \dot{\theta}_{\mathrm{w}_{\mathrm{left}}}\right)$$

$$\mathrm{z}_{\mathrm{w}} \coloneqq \mathrm{l}_{\mathrm{w}} \, , \; \mathrm{z}_0 \coloneqq 0 \, , \; \varphi \coloneqq \varphi_{\mathrm{b}} = \varphi_{\mathrm{w}}$$

$$\mathrm{E_K} \triangleq \mathrm{E_K^{body}} + \mathrm{E_K^{wheel}} \, , \; \mathrm{E_P} \triangleq \mathrm{E_P^{body}} + \mathrm{E_P^{wheel}} \, , \; \mathcal{L} \triangleq \mathrm{E_K} - \mathrm{E_P}$$

$$\begin{aligned} E_K = \frac{m_b}{2} \Big(& \frac{l_w^2}{4} \left(\dot{\theta}_{w_{left}} + \dot{\theta}_{w_{right}}\right)^2 + l_b^2 \left(\dot{\theta}_b^2 + \right. \\ & \left. + \frac{l_w}{L}^2 \left(\dot{\theta}_{w_{right}} - \dot{\theta}_{w_{left}}\right)^2 sin^2(\theta_b)\right) \Big) + \\ & + \frac{m_b \, l_b \, l_w}{2} \left(\dot{\theta}_{w_{left}} + \dot{\theta}_{w_{right}}\right) \dot{\theta}_b \, cos(\theta_b) + \frac{I_b}{2} \, \dot{\theta}_b^2 + \\ & + \frac{m_w \, l_w^2}{4} \left(\dot{\theta}_{w_{left}}^2 + \dot{\theta}_{w_{right}}^2\right)) + \\ & + \frac{I_w}{2} \left(\dot{\theta}_{w_{left}}^2 + \dot{\theta}_{w_{right}}^2\right) \end{aligned}$$

$$\mathrm{E_P} = \mathrm{m_b \, g \left(l_b \cos(\theta_b) + l_w\right) + m_w \, g \, l_w}$$

$$\begin{aligned} & \partial_{t \, , \, \dot{\theta}_b} \mathcal{L} - \partial_{\theta_b} \mathcal{L} = Q_{\theta_b} \Rightarrow \\ & m_b \, l_b^2 \, \ddot{\theta}_b - \frac{m_b \, l_b \, l_w}{2} \left(\ddot{\theta}_{w_{left}} + \ddot{\theta}_{w_{right}}\right) \dot{\theta}_b \, sin(\theta_b) + I_b \, \ddot{\theta}_b + \\ & - \frac{m_b \, l_b^2 \, l_w^2}{L^2} \left(\dot{\theta}_{w_{right}} - \dot{\theta}_{w_{left}}\right)^2 sin(\theta_b) \, cos(\theta_b) + \\ & + \frac{m_b \, l_b \, l_w}{2} \left(\dot{\theta}_{w_{left}} + \dot{\theta}_{w_{right}}\right) \dot{\theta}_b \, sin(\theta_b) - m_b \, g \, l_b \, sin(\theta_b) = Q_{\theta_b} \end{aligned}$$

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\begin{aligned}
&\partial_{t\,,\dot{\theta}_{w_{left}}}\mathcal{L} - \partial_{\theta_{w_{left}}}\mathcal{L} = Q_{\theta_{w_{left}}} \Rightarrow \\
&\quad \frac{m_b\, l_w^2}{4}\left((\ddot{\theta}_{w_{left}} + \ddot{\theta}_{w_{right}}) - \frac{l_b\, l_w^2}{L^2}\,(\ddot{\theta}_{w_{right}} - \ddot{\theta}_{w_{left}})\, sin^2(\theta_b)\right) + \\
&\quad - \frac{m_b\, l_b\, l_w}{2}\,(\dot{\theta}_{w_{left}} + \dot{\theta}_{w_{right}})\,\dot{\theta}_b\, cos(\theta_b) \\
&\quad + \frac{m_b\, l_b\, l_w}{2}\,\ddot{\theta}_{w_{left}}\, cos(\theta_b) + \frac{m_w\, l_w^2}{4}\,(\ddot{\theta}_{w_{left}} + \ddot{\theta}_{w_{right}}) + \\
&\quad + I_w\, \theta_{w_l eft}\,(\ddot{\theta}_{w_{left}} + \ddot{\theta}_{w_{right}}) = Q_{\theta_{w_{left}}}
\end{aligned}
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\begin{aligned}
&\partial_{t\,,\dot{\theta}_{w_{right}}}\mathcal{L} - \partial_{\theta_{w_{right}}}\mathcal{L} = Q_{\theta_{w_{right}}} \Rightarrow \\
&\quad \frac{m_b\, l_w^2}{4}\left((\ddot{\theta}_{w_{left}} + \ddot{\theta}_{w_{right}}) + \frac{l_b\, l_w^2}{L^2}\,(\ddot{\theta}_{w_{right}} - \ddot{\theta}_{w_{left}})\, sin^2(\theta_b)\right) + \\
&\quad + \frac{m_b\, l_b\, l_w}{2}\,\ddot{\theta}_{w_{right}}\, cos(\theta_b) + \frac{m_w\, l_w^2}{4}\,(\ddot{\theta}_{w_{left}} + \ddot{\theta}_{w_{right}}) + \\
&\quad + I_w\, \theta_{w_{right}}\,(\ddot{\theta}_{w_{left}} + \ddot{\theta}_{w_{right}}) = Q_{\theta_{w_{right}}}
\end{aligned}
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